



Fig. 6: Components soldered with IndusBoard

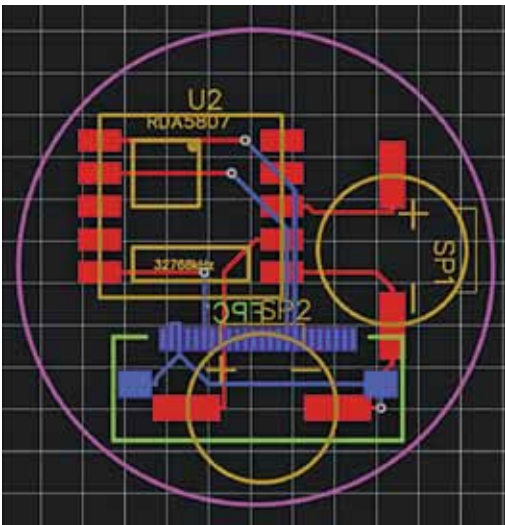


Fig. 7: PCB of the device

around to 108.0MHz

- The display updates to show the new frequency (for example, '87.4MHz')

- Test the next button:

- Tapping this button increases the frequency by 0.1MHz (for example, from 87.4MHz to 87.5MHz)

- If the frequency exceeds 108.0MHz, it wraps around to 87.5MHz

- The display updates accordingly

Play and pause using the middle button. The middle button is the play/pause button (at coordinates `playPauseButtonX = 120`, `playPauseButtonY = 150`), with a radius of 30 pixels and a gradient effect (magenta outer ring, purple inner ring).

- Test playback:
 - Tap the play/pause button.
 - If the radio is paused (`isPlaying = false`), the button shows a play icon (a triangle). Tapping it unmutes the RDA5807M and changes the icon to a pause symbol (two rectangles).
 - If the radio is playing (`isPlaying = true`), the button shows a pause icon. Tapping it mutes the RDA5807M and changes the icon to a play symbol.

- Audio output:
 - If the RDA5807M is detected (indicated by a green status dot), audio should be audible through the speaker connected to the LOUT/ROUT pins of the RDA5807M when the radio is unmuted.

- Tune to a known FM station (between 87.5 and 108.0MHz) using the previous/next buttons for clear audio output.

- If the RDA5807M is not detected (indicated by a red status dot), the device switches to simulation mode, updating the UI without generating audio.

Debugging and additional checks.

- Using the serial monitor:
 - Connect the IndusBoard to a computer via a USB-to-serial adaptor (using TX/RX pins, if available).
 - Check for messages like 'RDA5807M detected' or 'RDA5807M

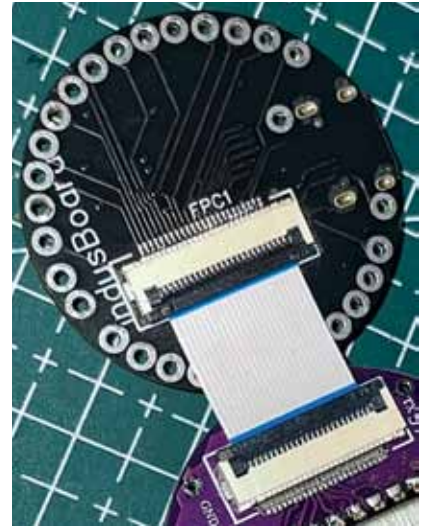


Fig. 8: IndusBoard with an FPC connector

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Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

not detected, running in simulation mode' to confirm the radio module's status.

- Monitor touch coordinates when tapping buttons (like 'Touch detected - Raw (x, y): (1234, 567) Mapped (x, y): (89, 234)').

- Touch accuracy:

- If touch buttons are unresponsive or inaccurate, adjust the coordinate mapping in the code accordingly.

The second version of FM radio that fits within an earbud will be published soon. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-designing-smallest-fm-radio>

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